6 Months Training TR-104 Full Stack Development

January, 2026

Day 2 – Angular Components, Interpolation, and Signals

1. Objective of the Day

The objective was to understand the core building blocks of Angular applications, including components, data binding, and modern state management using signals.

2. Tasks Performed

- Studied Angular's component-based architecture
- Explored structure of standalone Angular components
- Understood key files: app.ts, app.html, and app.css
- Implemented interpolation using {{ }} to display dynamic data
- Learned about selectors and how components are rendered in HTML
- Introduced signals for reactive state management
- Practiced updating signals using .set() method
- Implemented a counter example using signals
- Created input fields to capture user data
- Handled user input using (input) event
- Used $event and type casting for safe data extraction
- Implemented button click functionality using (click) event
- Created reset functionality using signals

3. Tools / Technologies Used

- Angular Framework (Latest Version)
- TypeScript
- HTML & CSS

4. Key Learnings

- Components are the fundamental building blocks of Angular
- Interpolation enables one-way data binding from logic to UI
- Signals provide reactive and predictable state management
- UI updates automatically when signal values change
- Event binding connects user interaction with application logic

5. Challenges Faced

- Understanding correct usage of signals and .set() method
- Handling event objects and extracting values properly

6. Solutions

- Practiced multiple examples for signal updates
- Used type casting (HTMLInputElement) for accurate data handling

7. Conclusion

- Successfully implemented dynamic UI updates using signals
- Built interactive components with user input and event handling